

Chromium-53

⁵³Cr (Cr I), nuclear spin $I = 3/2$. Strongest classified transitions below 2 μm (NIST): 56 levels, 87 lines.

Cr⁺ ionisation limit 54575.60 cm⁻¹ = 6.76651 eV (183.232 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element $| \langle U || er || J' \rangle |$ in ea₀
Matrix elements are measured or NIST values; * = high-accuracy theory, ≈ = model calculation.
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, $n = 70$.
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

